

System Architecture Document (SAD)

For

Nexora: Multi-Vendor E-Commerce Marketplace

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2. Document Information

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Project Title	Nexora Multi-Vendor E-Commerce Marketplace
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July 26, 2026	1.0.0	Chief Enterprise Solution Architect	Initial baseline system architecture document aligned

			with the Nexora SRS.
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5. Architecture Overview

Nexora is designed as an enterprise-grade, highly modular, scalable, multi-vendor e-commerce marketplace platform. The architectural blueprint follows a rigorous modular monolithic design pattern structured around clean architecture principles and strict separation of concerns. This ensures that while the initial delivery is streamlined via a unified deployable artifact, internal boundaries mirror microservice domain boundaries. Consequently, future migration to

containerized microservices orchestrated via Kubernetes can be accomplished with minimal structural friction.

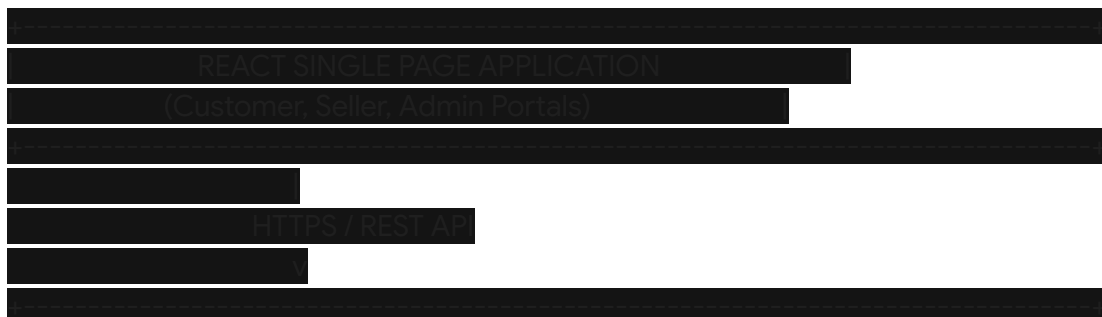
The platform bridges independent merchants and digital consumers by handling high-throughput catalog browsing, concurrent inventory reservations, multi-vendor cart splitting, secure payment settlements via asynchronous webhooks, and comprehensive administrative reporting.

6. Architectural Goals

- **Production Ready:** Designed to withstand high-volume retail traffic with robust error boundaries, graceful degradation, and containerized operational parity.
- **Enterprise Grade:** Incorporates enterprise security standards, immutable audit trails, centralized exception handling, and distributed caching.
- **Modular:** Enforces strict domain boundaries across functional modules (Authentication, Catalog, Order, Payment, Shipping, etc.).
- **Scalable:** Supports horizontal scaling of backend compute instances behind load balancers, optimized database indexing, and Redis caching layers.
- **Maintainable:** Built on clean code practices, SOLID principles, and layered architectures with clear component responsibilities.
- **Secure:** Implements stateless JWT authentication, RBAC, strict payload validation, and OWASP top-10 mitigation strategies.
- **Testable:** Facilitates isolated unit and integration testing via Mockito and JUnit with high code coverage baselines.
- **Cloud & Docker Ready:** Containerized using Docker for seamless orchestration across cloud environments.
- **Microservice Ready:** Domain logic is decoupled via event-driven messaging (Apache Kafka) to allow seamless extraction into independent services in the future.

7. High Level System Architecture

The end-to-end architecture spans from the client presentation tier down to persistent relational and caching layers.





8. Layered Architecture

Nexora enforces a strict multi-tiered layered architecture. Communication flows downward through the layers; upward traversal of dependencies is strictly prohibited.

- **Presentation Layer:** Represents the React single-page application handling user interfaces, state management, routing, and client-side validation across customer, seller, and administrator portals.
- **Controller Layer:** Exposes RESTful endpoints, handles HTTP protocol mapping, manages request parameter binding, enforces route-level security, and delegates payloads to the service layer without containing business logic.
- **Service Layer:** Encapsulates core business rules, transactional boundaries, workflow orchestration, event publishing, and domain logic validation.
- **Repository Layer:** Acts as the data access abstraction layer, managing data persistence operations, custom query definitions, and pagination specifications against the underlying data store.
- **Persistence Layer:** Comprises entity mapping definitions, database schema configurations, indexing rules, and object-relational mapping (ORM) session states managed via Hibernate.
- **Database Layer:** Provides persistent, ACID-compliant relational data storage via MySQL alongside high-performance in-memory caching and session storage via Redis.

9. Package Structure

The backend source code is organized into a clean, scalable package structure reflecting modular layered architecture:

```
com.nexora.marketplace
├──
├── config          # Global configuration classes (Security, Kafka, Redis, Swagger)
├── controller      # REST API endpoint controllers grouped by domain
├── service         # Business logic interface definitions
├── service.impl    # Concrete implementations of service interface
├── repository      # Spring Data JPA data access interfaces
├── entity          # JPA database entity models
├── dto.request     # Incoming data transfer request objects
├── dto.response    # Outgoing data transfer response objects
├── mapper          # Entity-to-DTO and DTO-to-Entity mapping components
├── security        # JWT filters, token providers, and security configuration
├── exception       # Custom domain exceptions and global exception handler
├── validation      # Custom Bean Validation annotations and validators
├── util           # Common utility and helper classes
├── common          # Shared constants, enums, and base classes
├── constant        # Application-wide constant definitions
├── cache           # Redis cache configuration and service components
├── event           # Kafka event producers, consumers, and event payloads
├── scheduler       # Scheduled background jobs (cleanup, reminders, reports)
├── audit           # Auditor awareness and immutable audit log components
```

10. Module Architecture

10.1 Authentication Module

- **Purpose:** Manages user identity verification, credential validation, and session security token issuance.
- **Responsibilities:** User login, JWT generation, token validation, password hashing, and token revocation upon logout.
- **Major Components:** Authentication filters, token provider, password encoder, security configuration.
- **Dependencies:** User Management Module, Security Core.

10.2 User Management Module

- **Purpose:** Handles user accounts, profile details, address books, and administrative user controls.
- **Responsibilities:** User registration, profile updates, address management, account activation/deactivation, and audit logging.
- **Major Components:** User controllers, profile services, address repositories, user entities.
- **Dependencies:** Authentication Module, Core Kernel.

10.3 Seller Management Module

- **Purpose:** Governs merchant onboarding, business verification, and storefront customization.
- **Responsibilities:** Seller application submission, administrative review workflows, storefront profile management, and status suspension.
- **Major Components:** Seller onboarding services, verification workflows, storefront controllers.
- **Dependencies:** User Management Module, Authentication Module.

10.4 Category Management Module

- **Purpose:** Manages hierarchical product category taxonomies.
- **Responsibilities:** Creating parent-child category trees, enabling/disabling categories, and assigning products to leaf nodes.
- **Major Components:** Category hierarchy services, category controllers, tree traversal utilities.
- **Dependencies:** Core Kernel.

10.5 Product Management Module

- **Purpose:** Handles merchant catalog listings and product configurations.
- **Responsibilities:** CRUD operations for simple and configurable products, product variant management, and high-resolution image asset linking.
- **Major Components:** Product catalog services, variant managers, image upload handlers.
- **Dependencies:** Category Management Module, Seller Management Module.

10.6 Inventory Management Module

- **Purpose:** Tracks real-time stock quantities and warehouse stock allocations.
- **Responsibilities:** Stock decrementing upon order confirmation, stock release on expiration, and low-stock threshold alerting.
- **Stock Allocation & Race Condition Mitigation Strategy:** To prevent race conditions during high-concurrency flash sales, inventory decrements utilize database-level optimistic locking coupled with Redis distributed locks. When an order checkout is initiated, a temporary Redis stock reservation is acquired with a strict TTL (Time-To-Live). If payment succeeds, stock is atomically deducted from MySQL using version-checked

optimistic locking (@Version). If payment expires or fails, the Redis reservation expires and stock remains untouched.

- **Major Components:** Inventory tracking services, low-stock trigger listeners, Redis lock managers.
- **Dependencies:** Product Management Module, Kafka Messaging.

10.7 Cart Module

- **Purpose:** Manages persistent shopping carts for authenticated customers.
- **Responsibilities:** Adding, updating, and removing cart items, validating stock/pricing on access, and merging guest carts upon login.
- **Major Components:** Cart session services, item validation components, cart repositories.
- **Dependencies:** Product Management Module, User Management Module.

10.8 Wishlist Module

- **Purpose:** Maintains customer product wishlists.
- **Responsibilities:** Saving preferred items, moving items from wishlist to cart, and triggering out-of-stock or price-drop notifications.
- **Major Components:** Wishlist services, item transfer handlers.
- **Dependencies:** Product Management Module, Customer Module.

10.9 Order Module

- **Purpose:** Manages order placement, state transitions, and cart splitting.
- **Responsibilities:** Converting carts into orders, splitting multi-vendor checkouts into sub-orders, tracking lifecycle states (PENDING, PROCESSING, SHIPPED, DELIVERED, CANCELLED, REFUNDED), and managing cancellations.
- **Major Components:** Order processing services, sub-order routers, state machine validators.
- **Dependencies:** Cart Module, Inventory Management Module, Payment Module, Shipping Module.

10.10 Payment Module

- **Purpose:** Integrates payment gateways and manages transaction ledgers.
- **Responsibilities:** Processing secure payments, handling asynchronous payment webhook callbacks, logging gateway responses, and managing refunds.
- **Major Components:** Gateway integration adapters, webhook signature verification services, refund managers.
- **Dependencies:** Order Module, Kafka Messaging.

10.11 Shipping Module

- **Purpose:** Calculates freight charges and tracks shipments.
- **Responsibilities:** Evaluating seller shipping rules, calculating delivery fees during

checkout, capturing courier tracking numbers, and providing milestone updates.

- **Major Components:** Shipping rate calculators, courier webhook handlers, tracking services.
- **Dependencies:** Order Module, Seller Management Module.

10.12 Review Module

- **Purpose:** Manages customer feedback and product ratings.
- **Responsibilities:** Capturing verified purchase star ratings and textual reviews, handling seller replies, and moderating inappropriate content.
- **Major Components:** Review submission services, moderation controllers, rating aggregators.
- **Dependencies:** Order Module, Product Management Module.

10.13 Notification Module

- **Purpose:** Dispatches multi-channel communications.
- **Responsibilities:** Sending transactional emails, SMS notifications, and managing the in-app notification center.
- **Major Components:** Email dispatchers, notification event listeners, template engines.
- **Dependencies:** Kafka Messaging, Core Kernel.

10.14 Dashboard Module

- **Purpose:** Renders role-specific operational metrics and charts.
- **Responsibilities:** Summarizing sales volume for sellers, pending tasks for administrators, and recent orders for customers.
- **Major Components:** Dashboard aggregation services, metric calculators.
- **Dependencies:** Order Module, Analytics Module.

10.15 Reports Module

- **Purpose:** Generates exportable business intelligence summaries.
- **Responsibilities:** Exporting financial sales reports, tax summaries, and inventory logs in CSV and PDF formats.
- **Major Components:** Report generation engines, export formatters, scheduling services.
- **Dependencies:** Analytics Module, Order Module.

11. Request Flow

The complete request lifecycle for an authenticated mutating operation (e.g., placing an order) traverses the architecture as follows:



12. Authentication & Authorization Architecture

- **JWT Flow:** Upon successful authentication of user credentials, the system issues a short-lived JSON Web Token containing subject claims, roles, and permissions. Subsequent client requests include this token in the Authorization: Bearer <token> HTTP header.
- **Spring Security Flow:** A custom stateless security filter intercepts incoming requests, extracts the JWT, validates its cryptographic signature and expiration, constructs an authenticated security context, and populates the Spring Security Context Holder.
- **Role-Based Authorization:** Method-level security annotations enforce strict role-based access control (ROLE_CUSTOMER, ROLE_SELLER, ROLE_ADMIN) across service and controller boundaries.
- **Token Lifecycle:** Access tokens expire after 15 minutes. A companion refresh token

mechanism allows clients to request new access tokens without re-entering credentials.

- **Password Encryption Strategy:** Passwords are never stored in plaintext. They are salted and hashed using the bcrypt cryptographic algorithm with a work factor of 12.

13. Exception Handling Strategy

- **Global Exception Handling:** A centralized exception handling component intercepts all unhandled runtime and checked exceptions across the application lifecycle.
- **Validation Errors:** Payload validation failures return an HTTP 400 Bad Request accompanied by a structured list of field-level violation messages.
- **Business Exceptions:** Custom business exceptions (e.g., `InsufficientStockException`, `SellerNotApprovedException`) map directly to appropriate HTTP status codes (409 Conflict, 403 Forbidden).
- **Database Exceptions:** Data integrity violations (e.g., unique constraint violations on email or SKU) are caught and translated into user-friendly client responses without leaking internal stack traces.
- **Authentication Exceptions:** Expired or invalid tokens trigger HTTP 401 Unauthorized responses.
- **API Error Response Format:** All error payloads adhere to a standardized JSON schema containing timestamp, HTTP status code, error code, error message, and request path.

14. Validation Strategy

- **Bean Validation:** Incoming request payloads are validated declaratively using standard annotations (`@NotNull`, `@Size`, `@Email`, `@Min`, `@Max`) at the controller boundary.
- **Custom Validation:** Domain-specific constraints (e.g., validating that promotional discount prices do not exceed original list prices) are implemented via custom validator annotations.
- **Input Validation:** Sanitization filters strip malicious HTML and script tags from string inputs to prevent Cross-Site Scripting (XSS).
- **Business Validation:** Complex cross-entity validations (e.g., verifying seller status prior to product creation) are executed within the service layer.

15. Logging Strategy

- **Application Logs:** Structured JSON logs capture standard application lifecycle events, incoming requests, execution durations, and operational state transitions.
- **Security Logs:** Dedicated security logs record all authentication attempts, token issuances, failed logins, and access control violations for forensic auditing.
- **Audit Logs:** Immutable audit logs record all administrative actions, data modifications, and state updates with actor identification and timestamps.
- **Error Logs:** Stack traces and unhandled exceptions are logged at the ERROR level with correlation IDs to facilitate rapid debugging.
- **Dynamic Log Levels:** Log levels are dynamically configurable via centralized management endpoints without requiring application restarts.

16. Caching Strategy

- **Redis Integration:** An in-memory Redis data store acts as a distributed cache layer for high-frequency read operations and session data management.
- **What Should Be Cached:** * Product catalog browsing categories and hierarchical tree structures.
 - Frequently accessed product details and variant attributes.
 - Static platform configuration parameters and tax rule lookup tables.
- **TTL Strategy:** Product catalog entries utilize a Time-To-Live (TTL) of 30 minutes. Category hierarchies utilize a TTL of 2 hours.
- **Cache Eviction Strategy:** Cache invalidation and eviction occur asynchronously via event listeners published to Kafka whenever a seller updates product details or administrators modify category taxonomies.

17. Messaging Architecture

- **Apache Kafka:** Asynchronous event-driven messaging is implemented using Apache Kafka to decouple core transactional workflows and ensure high throughput.
- **Event Flow Architecture:**
 - **Order Created:** Published when a checkout completes successfully; triggers inventory decrements, cart clearance, and payment confirmation hooks.
 - **Payment Completed:** Published upon gateway webhook verification; updates order status to PROCESSING and notifies the fulfillment engine.
 - **Inventory Updated:** Published when stock thresholds fall below 5 units; triggers low-stock alerts to sellers.
 - **Notification Sent:** Consumed by the notification service to dispatch transactional emails and SMS alerts.
 - **Seller Approved:** Published when an administrator approves a merchant application; provisions storefront access and initializes merchant ledgers.

18. API Architecture

- **REST Principles:** APIs adhere to Richardson Maturity Model Level 3 REST principles utilizing standard HTTP methods (GET, POST, PUT, PATCH, DELETE).
- **Versioning Strategy:** URI path versioning (/api/v1/...) is enforced across all endpoints to ensure backward compatibility during client upgrades.
- **Response Structure:** All API responses follow a uniform envelope structure containing status indicators, metadata, and payload data.
- **HTTP Status Codes:** Standard HTTP status codes are utilized strictly (200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 409 Conflict, 500 Internal Server Error).
- **Pagination, Sorting, and Filtering:** List endpoints support standardized pagination parameters (page, size), multi-field sorting (sortBy, direction), and dynamic attribute filtering.

19. Database Access Strategy

- **Spring Data JPA & Hibernate:** Relational data persistence is managed through Spring Data JPA repositories abstracting Hibernate ORM implementations.

- **Transactions:** Declarative transaction management (@Transactional) ensures ACID compliance across complex multi-table operations.
- **Loading Strategies:** Fetch types are explicitly defined as LAZY for collection associations to prevent N+1 select performance anti-patterns, with targeted JOIN FETCH queries utilized where eager loading is required.
- **Optimistic Locking:** Entity versioning (@Version) is applied to high-contention resources such as inventory stock levels to prevent lost updates during concurrent modifications.

20. Security Best Practices

- **OWASP Top 10 Mitigation:** Comprehensive safeguards protect against injection flaws, broken authentication, sensitive data exposure, and security misconfigurations.
- **SQL Injection Prevention:** Parameterized queries and JPA criteria builders completely eliminate raw SQL concatenation risks.
- **XSS Prevention:** Input sanitization and strict Content Security Policies (CSP) neutralize Cross-Site Scripting vectors.
- **CSRF Protection:** Stateless JWT authentication eliminates the vulnerability vector for Cross-Site Request Forgery on REST endpoints.
- **Rate Limiting:** API rate limiting is enforced at the gateway layer using Redis sliding window algorithms to prevent denial-of-service and brute-force attacks.

21. Design Patterns

- **Dependency Injection:** Leveraged extensively through Spring IoC container for loose coupling and testability.
- **Repository Pattern:** Abstracted data access layers separating business logic from persistence mechanisms.
- **DTO Pattern:** Encapsulates and isolates internal domain models from external API request/response payloads.
- **Builder Pattern:** Simplifies complex object construction, particularly for test fixture generation and DTO mapping.
- **Factory Pattern:** Manages dynamic instantiation of payment gateway adapters and shipping calculation engines.
- **Strategy Pattern:** Interchangeable calculation algorithms for shipping rates, tax computations, and commission splits.
- **Observer Pattern:** Event-driven decoupling via Spring Application Events and Kafka message publishing.
- **Singleton Pattern:** Enforced bean scopes across service and configuration components.

22. Scalability Strategy

- **Horizontal Scaling:** Backend Spring Boot application instances are stateless, allowing seamless horizontal scaling behind an NGINX load balancer.
- **Vertical Scaling:** Compute container resource allocations can be dynamically adjusted upward based on CPU and memory utilization metrics.
- **Load Balancing:** NGINX distributes incoming HTTP traffic evenly across available backend compute nodes using round-robin and least-connection algorithms.

- **Future Microservices:** Domain boundaries defined within the modular monolith allow future extraction of distinct domains (e.g., Order Service, Payment Service) into independent microservices communicating via Kafka.

23. Deployment Architecture

- **Docker:** The backend application and React frontend are fully containerized using multi-stage Dockerfiles to ensure environment parity.
- **Docker Compose:** Multi-container orchestration coordinates local and staging deployments encompassing the Spring Boot application, MySQL database, Redis cache, and Apache Kafka broker.
- **Cloud Deployment:** Production environments deploy container instances onto managed container services (e.g., AWS ECS or Kubernetes clusters) backed by managed relational databases (AWS RDS MySQL) and managed caching clusters (AWS ElastiCache Redis).
- **Environment Configuration:** Externalized configuration properties are injected securely via environment variables and Kubernetes ConfigMaps/Secrets.

24. Monitoring Strategy

- **Health Checks:** Spring Boot Actuator exposes health check endpoints (/actuator/health) monitoring database connectivity, disk space, memory utilization, and Redis availability.
- **Application Metrics:** Prometheus metrics collection tracks CPU usage, memory consumption, HTTP request throughput, and error rates.
- **Logging:** Centralized log aggregation streams structured JSON logs to observability platforms for real-time analysis and alerting.
- **Monitoring Dashboards:** Visual dashboards track system performance indicators and trigger automated alerts when operational thresholds are breached.

25. Testing Strategy

- **Unit Testing:** Business logic components, domain validators, and utility classes are tested in isolation using JUnit 5 and Mockito.
- **Integration Testing:** Repository layers and database interactions are validated using Spring Boot Test containers and embedded MySQL instances.
- **Controller Testing:** REST API endpoints are tested using MockMvc to verify request routing, payload validation, and HTTP response structures.
- **Test Coverage:** Code coverage tools enforce a strict minimum threshold of 80% unit test coverage across all business logic modules prior to build pipeline approval.

26. Coding Standards

- **Naming Conventions:** Classes follow PascalCase (OrderService), methods and variables follow camelCase (calculateSubtotal), and constants follow UPPER_SNAKE_CASE (MAX_LOGIN_ATTEMPTS).
- **Package Naming:** All packages are lowercase and structured hierarchically under com.nexora.marketplace.
- **Code Style:** Strict adherence to standard Java coding conventions, clean code principles, and SOLID design rules.

- **Documentation Standards:** Public APIs must be fully documented using Swagger/OpenAPI annotations, and complex business logic must include clear inline documentation explaining architectural intent.

27. Development Workflow

- **Git Branch Strategy:** Git Flow methodology is enforced, utilizing main for production releases, develop for integration, feature branches (feature/module-name), and bugfix branches.
- **Commit Standards:** Commit messages follow conventional commit guidelines (feat:, fix:, refactor:, test:).
- **Code Review:** Pull requests require mandatory peer code reviews and automated CI/CD pipeline verification before merging into develop.
- **Deployment Pipeline:** GitHub Actions automates building, running unit tests, scanning for security vulnerabilities, building Docker images, and pushing artifacts to container registries.

28. Risks & Mitigation

Risk Identified	Impact	Probability	Mitigation Strategy
Database Connection Exhaustion	High	Medium	Implement HikariCP connection pooling with optimized maximum pool sizes and connection timeout limits.
Inventory Overselling During Flash Sales	High	High	Utilize Redis distributed locks combined with database optimistic locking (@Version) for stock decrements.
Payment Webhook Failure or Duplication	High	Medium	Enforce idempotency keys on payment transactions and cryptographically verify webhook

			signatures.
Token Compromise / Session Hijacking	High	Low	Issue short-lived access tokens (15 mins), enforce secure HTTP-only cookies for refresh tokens, and support remote logout.
Monolithic Codebase Coupling	Medium	Medium	Strictly enforce modular package boundaries, preventing cross-domain repository calls without service-layer mediation.

29. Future Architecture Enhancements

1. Migration of modular monolithic backend architecture into fully containerized microservices orchestrated via Kubernetes.
2. Implementation of API Gateway pattern (Spring Cloud Gateway) for advanced routing, rate limiting, and cross-cutting security filtering.
3. Integration of distributed tracing (Zipkin / OpenTelemetry) across asynchronous Kafka event flows and REST calls.
4. Adoption of CQRS (Command Query Responsibility Segregation) pattern to optimize high-throughput read-heavy product catalog queries.
5. Integration of a distributed search engine cluster (Elasticsearch) for lightning-fast full-text product searching and faceted filtering.
6. Implementation of database read-replicas to scale read query performance independently from write operations.
7. Adoption of service mesh architecture (Istio) for secure service-to-service communication, mutual TLS, and traffic management.
8. Implementation of centralized distributed configuration management (Spring Cloud Config Server).
9. Integration of automated database migration management tools (Flyway) for robust schema version control across environments.
10. Adoption of reactive programming paradigms (Spring WebFlux) for high-concurrency

non-blocking I/O operations.

11. Integration of distributed circuit breaker patterns (Resilience4j) to prevent cascading failures across external service dependencies.
12. Implementation of multi-region database replication for global disaster recovery and low-latency geo-routing.
13. Adoption of event-sourcing patterns for immutable financial audit ledgers and order state history.
14. Integration of distributed session management across multi-node deployments using Redis session clustering.
15. Implementation of automated canary deployments and blue-green deployment pipelines via Kubernetes ingress controllers.
16. Adoption of gRPC protocols for high-performance internal inter-service communication upon microservice migration.
17. Integration of enterprise secrets management vaults (HashiCorp Vault) for secure credential storage.
18. Implementation of advanced automated chaos engineering testing suites to verify resilience under simulated network partitions.
19. Adoption of headless commerce architecture supporting decoupled omnichannel storefront endpoints.
20. Integration of distributed content delivery networks (CDNs) for static product image asset caching and global acceleration.
21. Implementation of real-time WebSocket communication channels for live customer chat and instant order status updates.
22. Adoption of automated API contract testing frameworks (Spring Cloud Contract) for microservice compatibility verification.
23. Integration of centralized application performance monitoring (APM) agents for deep execution profiling.
24. Implementation of automated database indexing tuning based on query execution plan analytics.
25. Adoption of immutable infrastructure provisioning using Infrastructure as Code (Terraform).
26. Integration of AI-driven anomaly detection models for automated fraudulent transaction identification.
27. Implementation of multi-tenant database schema isolation strategies for enterprise-grade data privacy.
28. Adoption of zero-trust network architecture principles across internal cluster communications.
29. Integration of automated web accessibility and performance regression testing in CI/CD pipelines.
30. Establishment of a multi-region active-active disaster recovery topology ensuring near-zero RTO and RPO metrics.

30. Appendix

Appendix A: Reference Standards

- Enterprise Integration Patterns (Hohpe & Woolf).
- Clean Architecture: A Craftsman's Guide to Software Structure and Design (Robert C. Martin).
- OpenAPI Specification 3.0 for Enterprise REST API Documentation.
- OWASP Application Security Verification Standard (ASVS).

Appendix B: Architecture Sign-Off

- Chief Enterprise Solution Architect: _____ Date: _____
- Principal Backend Engineer: _____ Date: _____
- Head of DevOps & Infrastructure: _____ Date: _____